

The Lagrange Top: Cubic Nutation, Two-Phase Reconstruction, and Exact Closure

Source-local theorem reconstruction

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Abstract

For the positive-inertia Lagrange symmetric heavy top we close the regular reduction–reconstruction problem. The inclination is inverted by one Jacobi function, both cyclic phases are complete third-kind elliptic integrals, and the full configuration closes exactly when two normalized phase increments are rational. Pole and repeated-root degenerations are kept outside singular Euler formulas. Exact computation certifies conventions, not the continuum proof.

1 Frozen convention and reduced cubic

Let $A = I_1 = I_2 > 0$, $C = I_3 > 0$, and $\gamma > 0$. In regular z - y - z Euler angles put $u = \cos \theta$, $L = p_\phi$, and $G = p_\psi$. Our convention is

$$H = \frac{p_\theta^2}{2A} + \frac{(L - Gu)^2}{2A(1 - u^2)} + \frac{G^2}{2C} + \gamma u. \quad (1)$$

The two momenta and $E = H$ are conserved. Since $p_\theta = A\dot{\theta}$, energy gives the reduced cubic owner

$$A^2 \dot{u}^2 = P(u) := 2A \left(E - \frac{G^2}{2C} - \gamma u \right) (1 - u^2) - (L - Gu)^2. \quad (2)$$

In particular

$$P(1) = -(L - G)^2, \quad P(-1) = -(L + G)^2. \quad (3)$$

The smooth Hamiltonian flow is complete: its potential is bounded on compact $\text{SO}(3)$, and positive kinetic energy makes each energy sublevel compact.

Theorem 1 (regular reduction). *A nonsteady regular inclination has finite period if and only if its allowed component of $P \geq 0$ is bounded by two simple roots in $(-1, 1)$. In the pole-incompatible chamber $L \neq \pm G$, label the roots*

$$-1 < r_1 < r_2 < 1 < r_3, \quad P = 2A\gamma(u - r_1)(u - r_2)(u - r_3).$$

This is the complete root classification for the regular reduced motion. The formulas below give its least reduced period, reconstruction, and full closure test.

Proof of the reduction. Allowed positions are exactly $P \geq 0$. At a simple turning root the time integral is locally $\int s^{-1/2} ds$, while a double root gives $\int s^{-1} ds$. Equation (3), the positive leading coefficient, and the cubic sign pattern give the stated ordering. This also separates finite nutation from a separatrix.

2 Elliptic inversion and two phase increments

Set

$$d = r_2 - r_1, \quad R = r_3 - r_1, \quad k^2 = \frac{d}{R}, \quad \nu^2 = \frac{\gamma R}{2A}, \quad Q_0 = \sqrt{2A\gamma R}.$$

Direct substitution proves

$$u(t) = r_1 + d \operatorname{sn}^2(\nu(t - t_0), k), \quad T = \frac{4A}{Q_0} K(k). \quad (4)$$

Indeed $z = (u - r_1)/d$ obeys $A^2 d^2 \dot{z}^2 = 2A\gamma d^2 R z(1 - z)(1 - k^2 z)$.

Hamilton's equations are

$$\dot{\phi} = \frac{L - Gu}{A(1 - u^2)}, \quad \dot{\psi} = \frac{G}{C} - u\dot{\phi}.$$

Define the finite complete integrals

$$I_N = \frac{2}{Q_0(1 - r_1)} \Pi \left(\frac{d}{1 - r_1}, k \right), \quad I_S = \frac{2}{Q_0(1 + r_1)} \Pi \left(-\frac{d}{1 + r_1}, k \right).$$

Partial fractions on the outbound and return legs give the two phase reconstruction

$$\Delta\phi = (L - G)I_N + (L + G)I_S, \quad (5)$$

$$\Delta\psi = G(1/C - 1/A)T + (G - L)I_N + (G + L)I_S. \quad (6)$$

Theorem 2 (full regular closure). *The full phase state in $T^*\mathrm{SO}(3)$ closes if and only if*

$$\frac{\Delta\phi}{2\pi} \in \mathbb{Q}, \quad \frac{\Delta\psi}{2\pi} \in \mathbb{Q}.$$

If their reduced denominators are q_ϕ, q_ψ , the least number of nutations in a full return is $\mathrm{lcm}(q_\phi, q_\psi)$.

Proof. After one least return of $(u, \dot{u})$, autonomy makes the two angle increments constant. Rational increments therefore suffice. Conversely, a full return must occur after an integer number of reduced returns. For $0 < \theta < \pi$, the z - y - z decomposition is unique modulo independent 2π shifts of ϕ, ψ , so both increments must be rational multiples of 2π . $\square$

3 Degenerate faces and quantum boundary

If $P(u_0) = P'(u_0) = 0$, initial data at u_0 give steady precession with two constant angular rates; its regular closure is their rational-ratio test. A nonconstant branch approaching the same double root takes infinite time. A triple root is another critical face and is not inserted into (4).

Equation (3) makes $L = G$ necessary at the north pole and $L = -G$ necessary at the south pole. Euler angles are singular there, so existence and uniqueness come from the smooth group Hamiltonian. If $L = G$ and two turning roots remain interior, formulas (4)–(6) have the finite limiting root $r_3 = 1$; an orbit that reaches the pole is reconstructed in a group chart, not by dividing by $1 - u^2$. Sleeping tops, $G = 0$, $\gamma = 0$ (the quadratic free-top boundary), and $A = C$ are separate faces.

The positive rigid-body metric yields a symmetric elliptic kinetic operator on $C^\infty(\mathrm{SO}(3))$. Elliptic completeness on the closed manifold makes it essentially self-adjoint; bounded real multiplication by γu preserves that property. Its unique closure equals the Friedrichs operator and has compact resolvent. We claim no closed spectrum.

4 Evidence, collision boundary, and scope

The executable artifact checks twelve rational parameter rows, sixty reconstruction probes, exact Sturm isolators, four Jacobi substitutions, and the steady/pole faces. A producer-independent checker and a symbolic lane own every stored field; repaired-hash mutations test the schema. These are finite convention receipts, not an enumeration proof.

C186 is the gravity-free Euler top; C244 has no body-axis spin momentum; C344 has a different complex-amplitude Poisson owner; C349 is a holonomic sphere oscillator. Thus this theorem owns the rigid-body two-phase reconstruction. The Route-A tuple is

$$(A0_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION),$$

and the overall verdict is rejection. Scope is `NO_BAD_EULER_OR_ROOT_NUMBER`; Route B is false. No target local data, Euler factor, root number, automorphy, target divisor or functional equation, target-zero match, or Hilbert–Pólya operator is asserted.

5 Sources

Cushman and Bates give an authoritative global Lagrange-top and reduction lineage [1]; Audin supplies an authoritative spinning-top integrability source [2]. The formulas and software here are independently reconstructed, without a priority claim.

References

- [1] R. Cushman and L. Bates, *Global Aspects of Classical Integrable Systems*, 2nd ed., Birkhäuser, 2015. DOI: 10.1007/978-3-0348-0918-4.
- [2] M. Audin, *Spinning Tops: A Course on Integrable Systems*, Cambridge University Press, 1996, ISBN 978-0-521-56129-7.